

Principles of Economics

Problem Set 12

Universidad Carlos III de Madrid | Department of Economics

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1. WARMING-UP QUESTIONS

1. What is inflation?
2. What does inflation refer to?
 - a) The price level is falling
 - b) The price level is rising
 - c) The inflation rate is falling
 - d) The inflation rate is rising
3. Which term describes a situation where the inflation rate is falling?
 - a) Inflation
 - b) Deflation
 - c) Disinflation
 - d) Stagflation
4. What is deflation and why is it important to avoid it?
5. What is the term for a general decrease in the price level?
 - a) Inflation
 - b) Deflation
 - c) Disinflation
 - d) Hyperinflation
6. If the price level is rising, which of the following is occurring?
 - a) Inflation
 - b) Deflation
 - c) Disinflation
 - d) None of the above
7. During a period of deflation, what happens to the general price level?
 - a) It rises
 - b) It remains constant
 - c) It falls
 - d) It fluctuates rapidly

8. Which of the following correctly describes disinflation?
- a) The price level is rising more slowly
 - b) The price level is falling
 - c) The inflation rate is falling
 - d) Both b and c
9. What is price stability and why is it so important?
10. In the event of a financial crisis, would it be preferable for the government to stabilize the economy using fiscal or monetary policy?
11. What are the dangers of using fiscal policy?
12. When might the government have no choice but to use fiscal policy?

2. MULTIPLE-CHOICE QUESTIONS

13. Which of the following statements is correct?
- a) Money is the cash (coins and notes) used as the medium of exchange to purchase goods and services.
 - b) Bank money is the total money in the savers' deposit accounts at the bank.
 - c) Base money is broad money plus legal tender held by banks minus bank money.
 - d) Liquidity transformation occurs when the banks transform illiquid deposits into liquid loans.
14. Which of the following statements are correct regarding the principal-agent problem?
- a) A principal-agent problem exists in loans due to a positive possibility of the principal not being repaid.
 - b) The principal-agent problem can be resolved by writing a binding contract for the borrower to exert full effort.
 - c) One solution for the principal-agent problem in loans is for the borrower to provide equity.
 - d) The principal-agent problem leads to credit rationing in the loans market.
15. The Swedish central bank raises the interest rate. Which, if any, of the following statements about the exchange rate (a) and about the consequences of the central bank's decision (b) and (c) are true?
- a. If a depreciation of home's exchange rate increases its value, then the exchange rate must be defined as the number of units of home currency per one unit of foreign currency.
 - b. The demand for Swedish bonds goes up and hence, the Swedish Krona depreciates in value.
 - c. Net exports, i.e. exports minus imports, rise because a Krona buys more units of foreign currency.

16. Which of the following statements regarding inflation and deflation is correct?
- Borrowers benefit from deflation, as the value of their debt decreases in real terms.
 - Inflation transfers wealth from lenders to borrowers.
 - Falling prices benefit consumers and are therefore always good for the economy.
 - Inflation makes it difficult for consumers and firms to attain the message about scarcity of resources (sent by relative prices) and is therefore always bad for the economy.
17. Consider a scenario where the Bank of England views the UK economy to be overheating and is attempting to slow the economy down using monetary policy. Which of the following statements regarding the effects of an interest rate rise is correct?
- It leads to higher bond prices, which results in higher demand for UK bonds.
 - It leads to higher demand for GBP, which results in an appreciation of the GBP.
 - It leads to the UK exports becoming cheaper and imports becoming more expensive.
 - It has opposing effects on the UK's aggregate demand (AD): it discourages investment, which lowers AD, but results in cheaper imports, which boosts AD.
18. The following diagram depicts the model of the labour market:



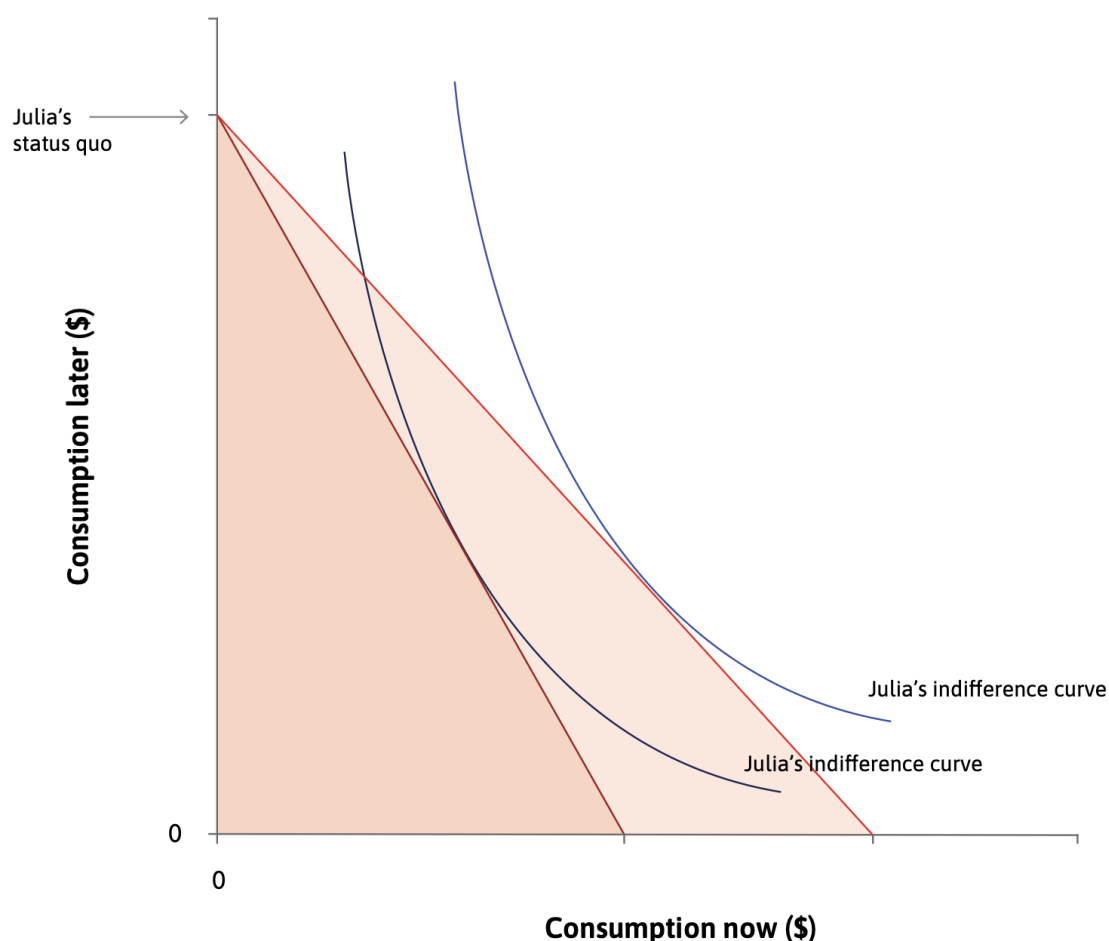
Suppose now that the government adopts policies that make it difficult for foreign firms to enter its markets. Assume that the level of employment and the labour supply remain constant. Which of the following statements regarding mechanisms by which inflation is created are correct?

- With reduced competition, firms can now charge a higher markup on their costs, raising the price-setting curve.

- b) With the labour market not in equilibrium at the lower real wage, the workers now lack the motivation to work at the given unemployment rate. Therefore, the wage is increased, described by an upward shift in the wage-setting curve.
- c) If the firms are able to continue charging the new higher markup, now applied to the new higher wage, the price rises again, lowering the real wage to the price-setting curve.
- d) After the price rise, if the workers are able to continue demanding the initial real wage as the minimum level required to motivate them to work, the wage rises again, increasing the real wage to the level on the wage-setting curve.

3. NUMERICAL PROBLEMS

19. The following diagram depicts Julia's choice of consumptions in periods 1 (now) and 2 (later) under different interest rates. She has no income in period 1 but an income in period 2 against which she can borrow. Based on this information, which of the following statements are correct?



- a) A cut in the interest rate increases the marginal rate of transformation of consumption from period 2 to period 1.
- b) Julia will unambiguously increase her consumption in period 1 after an interest rate cut.
- c) Julia will unambiguously decrease her consumption in period 2 after an interest rate cut.
- d) The graph of interest rate (vertical axis) versus period-1 consumption (horizontal axis) is

downward sloping.

- 20.** The following table shows the nominal interest rate and the annual inflation rate (the GDP deflator) of Japan in the period 1996–2015 (Source: World Bank).

	1996–2000	2001–2005	2006–2010	2011–2015
Interest rate	1.50%	1.40%	1.30%	1.20%
Inflation rate	–1.9%	–0.9%	–0.5%	1.60%

Based on this information, which of the following statements are correct?

- a) The real interest rate in 1996–2000 was 3.4%.
- b) Japan's real interest rate has been decreasing consistently over this period.
- c) Japan's real interest rate turned from being negative to positive during the period.
- d) The real interest rate has been increasing faster than the nominal interest rate.

4. CONCEPTUAL QUESTIONS

- 21.** Explain why a change in the central bank's policy interest rate affects the exchange rate through the market for financial assets (such as government bonds).